

An Optimum Loop Gain Tracking All-Digital PLL Using Autocorrelation of Bang–Bang Phase-Frequency Detection

Sungchun Jang, Sungwoo Kim, Sang-Hyeok Chu, Gyu-Seob Jeong, Yoonsoo Kim, *Student Member, IEEE*, and Deog-Kyoon Jeong, *Senior Member, IEEE*

Abstract—An all-digital phase-locked loop with a bang–bang phase-frequency detector (BBPFD) that tracks the optimum loop gain for minimum jitter is proposed. The autocorrelation of the output of BBPFD indicates whether the bang–bang PLL operates in the nonlinear regime or the random noise regime. An adaptive loop gain controller continuously evaluates the autocorrelation of the BBPFD output and adjusts the loop gain to make the autocorrelation zero. The digital loop filter operates at higher than the reference clock frequency to reduce the loop latency and to mitigate the resolution of the digitally controlled oscillator. The prototype chip has been fabricated in a 65-nm CMOS process. The core consumes 5 mW at 2.5 GHz and exhibits root-mean-square jitter of 1.72 ps.

Index Terms—Adaptive gain control, all-digital phase-locked loop (ADPLL), autocorrelation, bang–bang phase-frequency detector (BBPFD), bang–bang phase-locked loop (BBPLL).

I. INTRODUCTION

RECENTLY, all-digital phase-locked loops (ADPLLs) have been widely used in frequency synthesis for RF transceivers and in clock generation for system-on-a-chip applications [1]–[4]. ADPLLs offer many advantages in terms of area, scalability, programmability, and testability. One of the disadvantages of ADPLLs is in designing a high-resolution time-to-digital converter (TDC). A high-performance TDC usually consumes a lot of power and complicates the circuit design. A simple bang–bang phase-frequency detector (BBPFD) alleviates these issues. However, the strong nonlinearity of BBPFDs creates difficulties in using conventional analysis methods.

When the bang–bang quantization error is larger than the random noise fluctuation, limit cycles appear as large dithering jitter. The random noise found in building blocks can eliminate these limit cycles. However, a large random noise source also increases the random noise component of the output jitter. There is an optimum loop gain that minimizes the total output jitter when the jitter induced by quantization is equal to the

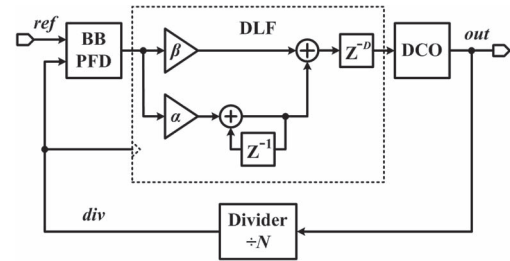


Fig. 1. Block diagram of a second-order all-digital BBPLL.

jitter induced by random noise [5]–[7]. However, the random noise variability due to process, voltage, and temperature (PVT) variation changes the optimum loop gain. Moreover, the gain of a digitally controlled oscillator (DCO) is also very sensitive to PVT variations. Therefore, the loop gain should be continuously calibrated during normal operation to minimize the output jitter.

This brief proposes an all-digital integer- N bang–bang PLL (BBPLL) that tracks the optimum loop gain and minimizes the output jitter. An adaptive loop gain controller (ALGC) adjusts the loop gain by estimating the autocorrelation of the BBPFD output. When the autocorrelation is zero, the output jitter is minimized. The conventional adaptive gain control method [8]–[10] provides a high loop gain before the ADPLL is locked in order to reduce the lock time. Then it also offers a low loop gain after the ADPLL is locked, thus reducing the dithering jitter. However, this method does not track the optimum gain point. The proposed gain control method increases the gain during frequency tracking to reduce the lock time and tracks the optimum gain to minimize the output jitter after the ADPLL is locked.

This brief is organized as follows. Section II describes the relationship between the output jitter and the autocorrelation of the BBPFD output. Section III details the circuit implementation of the proposed all-digital BBPLL. Section IV summarizes the measurement results, and Section V concludes this brief.

II. AUTOCORRELATION OF THE BBPFD OUTPUT

A conventional all-digital BBPLL architecture is shown in Fig. 1. The BBPFD provides an output of +1 when the divided clock leads the reference clock or an output of –1 when the reference clock leads the divided clock. To relate the BBPFD output characteristic and the BBPLL output jitter, the

Manuscript received December 10, 2014; revised March 4, 2015; accepted April 8, 2015. Date of publication May 20, 2015; date of current version August 28, 2015. This work was supported by IC Design Education Center (IDEC). This brief was recommended by Associate Editor S. Palermo.

The authors are with the Department of Electrical and Computer Engineering and the Inter-University Semiconductor Research Center (ISRC), Seoul National University, Seoul 151-742, Korea (e-mail: dkjeong@snu.ac.kr).

Color versions of one or more of the figures in this brief are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TCSII.2015.2435691

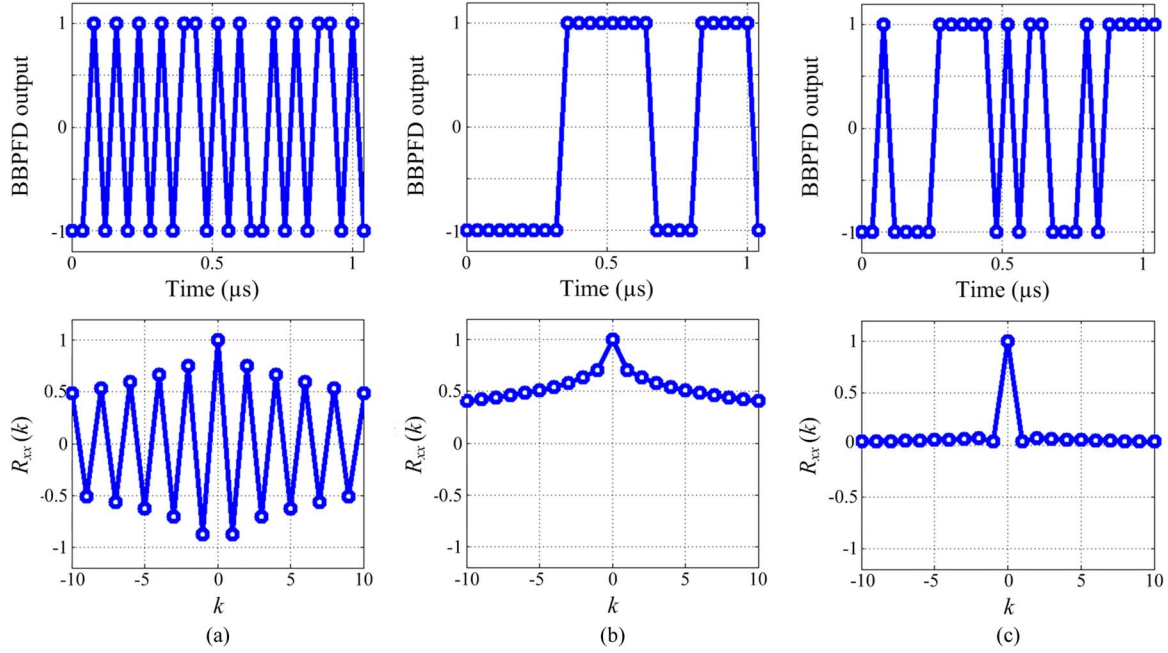


Fig. 2. Simulated BBPFD output and its autocorrelation using behavioral model. (a) High loop gain (nonlinear regime). (b) Low loop gain (random noise regime). (c) Optimum loop gain (minimum output jitter).

autocorrelation function of the BBPFD is used. The autocorrelation of a discrete time process is defined as

$$R_{xx}(k) = \frac{1}{(n-k)\sigma^2} \sum_{t=1}^{n-k} (x_t - \mu)(x_{t+k} - \mu) \quad (1)$$

where μ , σ , and n are the mean, the standard deviation, and the number of samples of a random variable x , respectively. When the BBPLL is locked, the average phase error is zero, and the BBPFD output is $+1$ or -1 . Therefore, $\mu = 0$ and $\sigma = 1$. The autocorrelation of the BBPFD output then is reduced to

$$R_{xx}(k) = \frac{1}{(n-k)} \sum_{t=1}^{n-k} x_t x_{t+k}. \quad (2)$$

Fig. 2 shows the simulated BBPFD output and its autocorrelation function in three different BBPLL operating regions when the loop delay D is zero. When the loop gain is high, the BBPLL operates in the nonlinear regime, and it exhibits a limit-cycle behavior, generating so-called hunting or dithering jitter. In this case, the BBPFD output shows trajectories with a period of 2 clock cycles, as shown in Fig. 2(a). In general, when there is no noise and frequency offset, it is periodic with a period of $4D + 2$ [11]. Owing to the random noise, the BBPFD output does not show the exact orbit of a period of 2, but the output mostly alternates between $+1$ and -1 . The autocorrelation $R_{xx}(k)$ is negative when k is an odd multiple of $2D + 1$ and positive when k is an even multiple of $2D + 1$. As k is increased, $R_{xx}(k)$ converges to zero owing to the effect of random noise. The BBPFD output is consecutive ups or downs when the loop gain is too small, as shown in Fig. 2(b). In the random noise regime, the gain is too small to track the phase error at a time, resulting in a slew-rate limited mode. Then, the BBPFD generates a burst of output bits of the same sign to track the phase error. Thus, the output shows a strong

positive relationship with neighboring samples; thus, $R_{xx}(k)$ is positive. However, as k is increased, $R_{xx}(k)$ converges to zero eventually. Fig. 2(c) shows the output of the BBPFD when the BBPLL operates in the minimum output jitter condition. In this region, the BBPFD outputs are uncorrelated, and $R_{xx}(k)$ is almost zero for any value of k except zero, and the noise can be modeled as white noise.

According to the autocorrelation characteristic, three operating regions can be defined, i.e.,

$$\begin{cases} -1 < R_{xx}(2D + 1) < 0 & \text{(Nonlinear regime)} \\ 0 < R_{xx}(2D + 1) < 1 & \text{(Random noise regime)} \\ R_{xx}(2D + 1) \approx 0 & \text{(Optimum point).} \end{cases}$$

By evaluating $R_{xx}(2D + 1)$, the BBPLL operating region is determined, and the loop gain is adjusted accordingly. The loop gain is decreased when $R_{xx}(2D + 1)$ is negative and increased when $R_{xx}(2D + 1)$ is positive.

We simulated the total output jitter and the autocorrelation of the BBPFD output $R_{xx}(2D + 1)$ as a function of βK_T using the behavioral model, where β is the proportional path gain of the digital loop filter (DLF) and K_T is the DCO gain. Comparisons are made between the simulated optimum gain points with the theoretical results presented in [5, eqs. (37) and (38)] that calculates the optimum bandwidth and minimum jitter under the given DCO phase noise. Fig. 3(a) shows the output jitter with three different DCO phase noise values: $\mathcal{L} = -90, -95$, and -100 dBc/Hz at 1 MHz offset when $D = 0$. As the DCO phase noise is decreased, the minimum output jitter and the optimum βK_T value are reduced. Fig. 3(b) shows $R_{xx}(1)$. As βK_T is increased, $R_{xx}(1)$ is decreased, and the BBPLL operating region moves from the random noise regime to the nonlinear regime. The βK_T value where $R_{xx}(1) = 0$ in Fig. 3(b) coincides with the βK_T value in Fig. 3(a), where

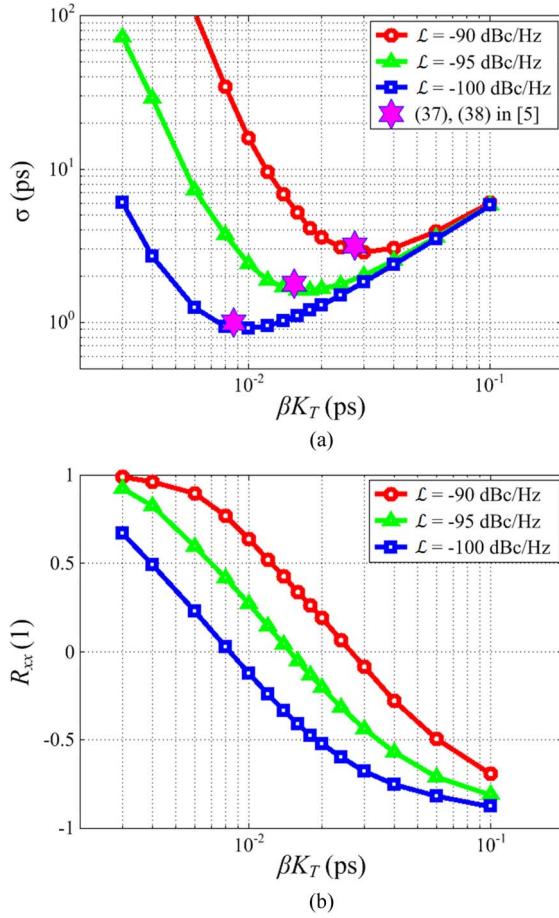


Fig. 3. Output jitter and autocorrelation of BBPFD output $R_{xx}(1)$ as a function of βK_T at three different DCO phase noise values $\mathcal{L} = -90$, -95 , and -100 dBc/Hz at 1 MHz offset. (a) Output jitter σ . (b) $R_{xx}(1)$. In this example, $\alpha = 2^{-10}$, $\beta = 0.05$, $f_c = 400$ kHz, $f_{\text{ref}} = 25$ MHz, $N = 100$, and $D = 0$.

the output jitter is minimized. These results are in excellent agreement with those in [5]. Fig. 4(a) shows the output jitter with three different loop delays: $D = 0, 1$, and 2 . As the loop delay is increased, the minimum output jitter is increased, and the optimum βK_T value is decreased. Fig. 4(b) shows that the output jitter is minimized when $R_{xx}(2D+1) = 0$.

III. CIRCUIT IMPLEMENTATIONS

The proposed digital BBPLL architecture is shown in Fig. 5. The ALGC controls the gain δ adaptively by estimating the autocorrelation of the BBPFD output. In order to have a large phase margin, β should be sufficiently higher than α , and the ratio α/β is fixed during the gain control. The 10-bit DCO consists of a three-stage pseudodifferential ring oscillator and a 32×32 current-source array. To increase the effective resolution of the DCO, a first-order delta-sigma modulator (DSM) with up to 20 fractional bits is added. The DSM operates at a higher frequency than the reference clock to increase the effective resolution of the DCO and to reduce the quantization noise. The DLF operates at the same clock rate as the DSM, which is the output of the dual-modulus prescaler. The high-frequency DLF clock reduces the loop latency and also reduces

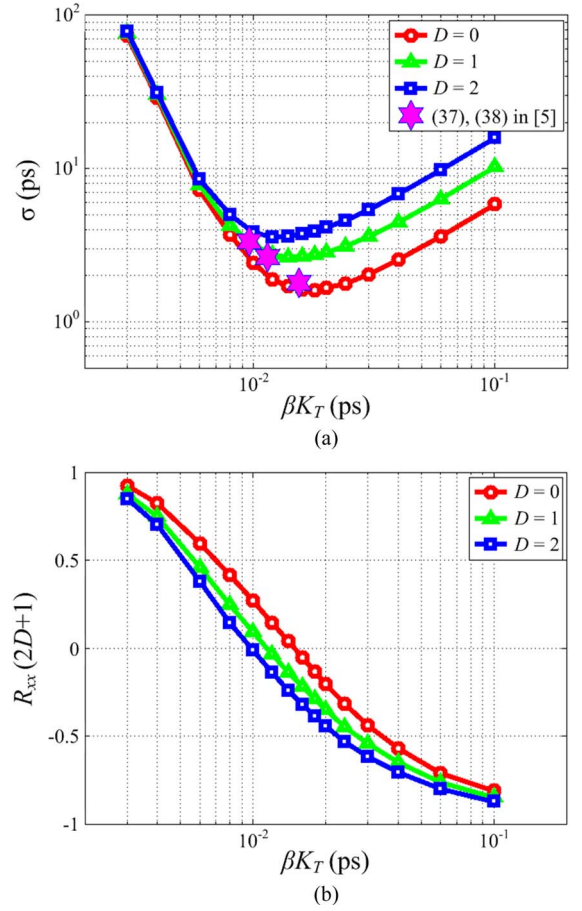


Fig. 4. Output jitter and autocorrelation of BBPFD output $R_{xx}(2D+1)$ as a function of βK_T at three different loop delays $D = 0, 1$, and 2 . (a) Output jitter σ . (b) $R_{xx}(2D+1)$. In this example, $\alpha = 2^{-10}$, $\beta = 0.05$, $\mathcal{L} = -95$ dBc/Hz at 1 MHz offset, $f_c = 400$ kHz, $f_{\text{ref}} = 25$ MHz, and $N = 100$.

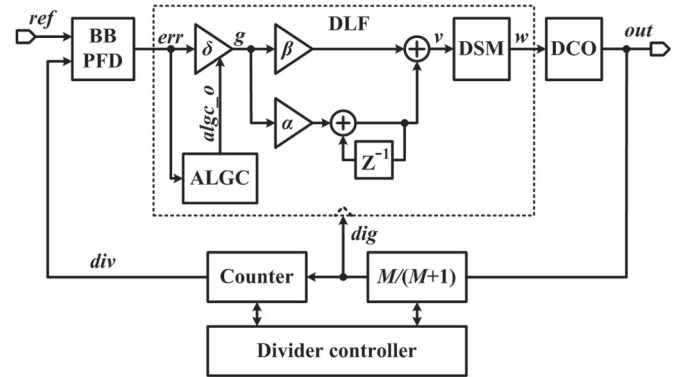


Fig. 5. Block diagram of the proposed digital BBPLL.

design complexity due to the clock domain crossing between the DLF and the DSM.

Fig. 6(a) shows the ALGC block diagram. The ALGC delays the BBPFD output by $(2D+1)$ cycle and multiplies the result by the current output. In our design, the latency is less than one reference clock cycle and $D = 0$. Then, the desired autocorrelation value R_{xx} is subtracted, and it is averaged out by the accumulator. The jitter is minimized when $R_{xx} = 0$. However, the R_{xx} value was externally varied just for the test purpose to confirm the relation between the amount of

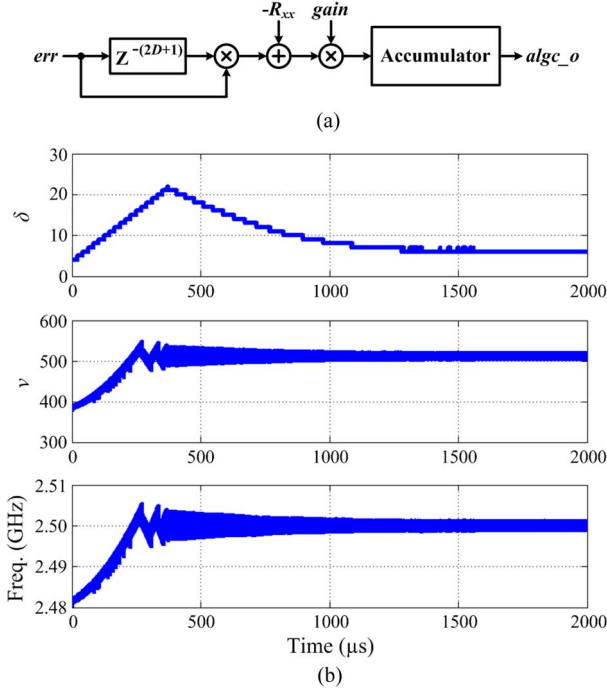


Fig. 6. (a) Block diagram of the ALGC. (b) Behavioral model transient simulation.

output jitter and various autocorrelation values. Fig. 6(b) shows the behavioral simulation waveforms. During the frequency tracking, the BBPFD output is either up or down for many cycles. The correlation between current and delayed BBPFD output is strongly positive. Then, the ALGC increases the loop gain. As the BBPLL approaches the lock state, the BBPFD output alternates between $+1$ and -1 . The current BBPFD output is negatively correlated with the delayed output. The loop gain is decreased by the ALGC and eventually settles when the autocorrelation is zero. The loop gain is high during frequency tracking and low after the PLL is locked as the conventional adaptive loop gain control method [8]–[10]. The proposed method reduces the lock time and also tracks the optimum loop gain.

Fig. 7 shows the timing diagram of the DLF. The DLF clock is generated from a dual-modulus prescaler that divides the DCO clock by M or $M + 1$. The DLF clock samples the reference clock and detects the rising edge of the reference clock. Then, the enable signal en shown in Fig. 7 is generated, which is high for one DLF clock period in each reference cycle. The error is $+1$ or -1 while en is high, and it is 0 while en is low. Then, the proportional path gain is reduced to $M\beta K_T$ or $(M + 1)\beta K_T$ depending on the divide value of the prescaler from $N\beta K_T$, where N is the total divide value. It thus mitigates the DCO resolution requirement. The total delay of the proportional path of the digital part is about four cycles of the digital clock, i.e., smaller than one cycle of the reference clock. The ALGC also uses the DLF clock, but it operates only when en is high.

IV. MEASUREMENTS

Fig. 8 shows a die photograph of the prototype chip that has been fabricated in a 65-nm CMOS process. The active area of

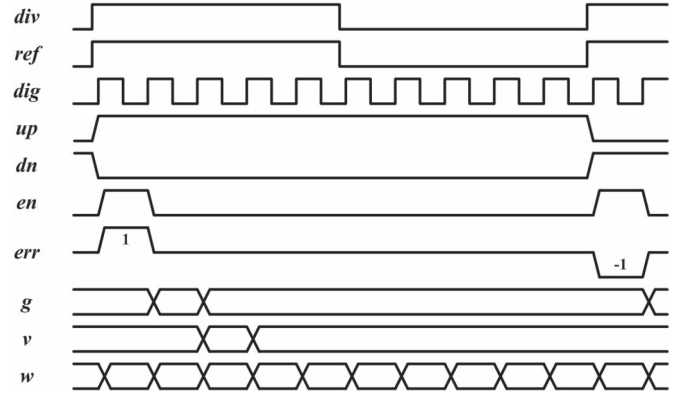


Fig. 7. Timing diagram of the DLF.

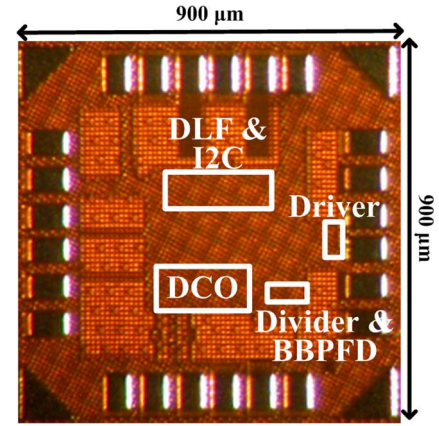


Fig. 8. Die photograph.

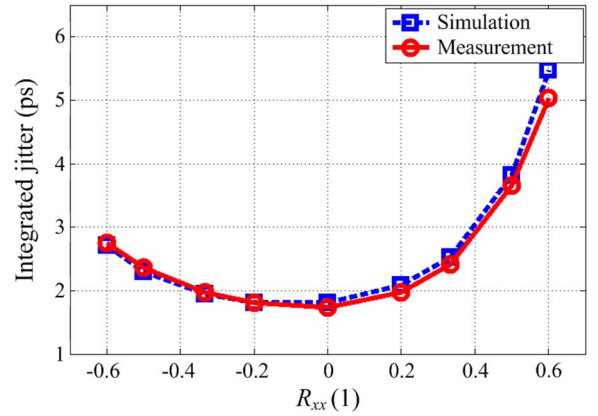


Fig. 9. Simulated and measured integrated jitter from 20 kHz to 20 MHz.

the circuit is 0.038 mm^2 . The area of the analog part, excluding the test driver, is 0.023 mm^2 , and the area of the digital part, including I2C, is 0.015 mm^2 .

Fig. 9 shows the measured integrated jitter from 20 kHz to 20 MHz at different settings of autocorrelation $R_{xx}(1)$. In this measurement, we used the loop parameters $\alpha = 2^{-8}$, $\beta = 1$, $f_{ref} = 25 \text{ MHz}$, and $N = 100$. In the behavioral model simulation, we used the DCO model parameters $\mathcal{L} = -95 \text{ dBc/Hz}$ at 1 MHz offset and corner frequency $f_c = 400 \text{ kHz}$. When the

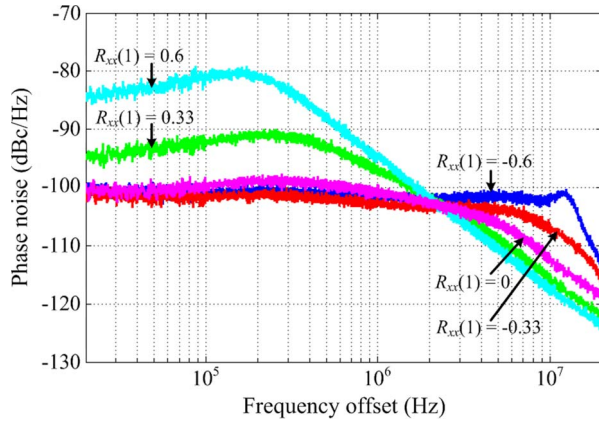


Fig. 10. Measured phase noise.

TABLE I
PERFORMANCE COMPARISON

	[2]	[3]	[8]	This Work
Technology (nm)	22	65	130	65
Supply (V)	1	1.2	1.2	1.2
Reference frequency (MHz)	40	500	50	25
Output frequency (GHz)	3.2	4	1.35	2.5
RMS jitter σ (ps)	3.1 ¹⁾	6 ²⁾	3.7	1.72
Area (mm ²)	0.017	0.03	0.2	0.038
Power (mW)	3.4	17.2	16.5	5
FOM ³⁾	-224	-212	-216	-228

1) Integrated jitter from 10 kHz to 500 MHz

2) Long term jitter

3) FOM = $10 \log \left[\left(\frac{\sigma}{1s} \right)^2 \cdot \frac{P}{1mW} \right]$

autocorrelation is zero, the integrated jitter is minimized. The measured jitter values are close to the simulation results. The measured root-mean-square (RMS) jitter is 1.72 ps.

Fig. 10 shows the measured phase noise with different settings of autocorrelation $R_{xx}(1)$. As the autocorrelation value is decreased, the spectral tones appear at half the reference frequency owing to the limit cycle. As the autocorrelation value is increased, the spectral peak disappears. However, a low loop gain reduces the loop bandwidth and increases the in-band noise. When the autocorrelation is zero, the spectrum has no spectral peak and also has low in-band noise.

Table I summarizes the performance of the proposed ADPLL and offers comparison with other works. The ADPLL dissipates 5 mW, excluding that dissipated by output driver. The power consumption of the analog and digital parts is 4.1 and 0.9 mW, respectively, at 2.5 GHz from a 1.2-V supply.

V. CONCLUSION

A digital BBPLL tracking the optimum loop gain to minimize the output jitter has been proposed. The ALGC adjusts the loop gain so that the autocorrelation of the BBPFD output over one cycle time distance becomes zero. It tracks the optimum loop gain over PVT variations with little hardware overhead.

REFERENCES

- [1] G. Marzin, S. Levantino, C. Samori, and A. L. Lacaita, "A 20 Mb/s phase modulator based on a 3.6 GHz digital PLL with -36 dB EVM at 5 mW power," *IEEE J. Solid-State Circuits*, vol. 47, no. 12, pp. 2974–2988, Dec. 2012.
- [2] N. August, H. Lee, M. Vandepas, and R. Parker, "A TDC-less ADPLL with 200-to-3200 MHz range and 3 mW power dissipation for mobile SoC clocking in 22 nm CMOS," in *Proc. IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2012, pp. 246–248.
- [3] J. A. Tierno, A. V. Rylyakov, and D. J. Friedman, "A wide power supply range, wide tuning range, all static CMOS all digital PLL in 65 nm SOI," *IEEE J. Solid-State Circuits*, vol. 43, no. 1, pp. 42–51, Jan. 2008.
- [4] A. Rylyakov *et al.*, "Bang-bang digital PLLs at 11 and 20 GHz with sub-200fs integrated jitter for high-speed serial communication applications," in *Proc. IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2009, pp. 94–95.
- [5] G. Marucci, S. Levantino, P. Maffezzoni, and C. Samori, "Analysis and design of low-jitter digital bang-bang phase-locked loops," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 61, no. 1, pp. 26–36, Jan. 2014.
- [6] M. Zanuso *et al.*, "Noise analysis and minimization in bang-bang digital PLLs," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 56, no. 11, pp. 835–839, Nov. 2009.
- [7] S. Tertinek, J. P. Gleeson, and O. Feely, "Statistical analysis of first-order bang-bang phase-locked loops using sign-dependent random-walk theory," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 57, no. 9, pp. 2367–2380, Sep. 2010.
- [8] D.-S. Kim, H. Song, T. Kim, S. Kim, and D.-K. Jeong, "A 0.3–1.4 GHz all-digital fractional-N PLL with adaptive loop gain controller," *IEEE J. Solid-State Circuits*, vol. 45, no. 11, pp. 2300–2311, Nov. 2010.
- [9] L. Xiu, W. Li, J. Meiners, and R. Padakanti, "A novel all-digital PLL with software adaptive filter," *IEEE J. Solid-State Circuits*, vol. 39, no. 3, pp. 476–483, Mar. 2004.
- [10] S.-Y. Yang, W.-Z. Chen, and T.-Y. Lu, "A 7.1 mW, 10 GHz all digital frequency synthesizer with dynamically reconfigured digital loop filter in 90 nm CMOS technology," *IEEE J. Solid-State Circuits*, vol. 45, no. 3, pp. 578–586, Mar. 2010.
- [11] N. Da Dalt, "A design-oriented study of the nonlinear dynamics of digital bang-bang PLLs," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 52, no. 1, pp. 21–31, Jan. 2005.